

Literature Review, Paper Summaries, and Search Strategies

For: Endogenous Education Choice in Segmented Search Markets

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Part I

Paper Summaries

1 Nekoei and Weber (2017) — “Does Extending Unemployment Benefits Improve Job Quality?”

Main point

More generous unemployment benefits improve the quality of post-unemployment matches. The standard view holds that longer UI raises moral hazard (workers search less), but the paper shows an offsetting “selectivity” channel: workers with more insurance can afford to be choosier and wait for better offers.

Specification

The authors exploit a sharp regression-discontinuity design in Austria. Workers who become unemployed after age 40 are entitled to 39 weeks of UI benefits; those just under 40 get only 30 weeks. This 9-week jump at an age threshold creates quasi-experimental variation in UI generosity that is unrelated to worker ability or match quality.

Key findings

- A 9-week UI extension increases reemployment wages by approximately 0.5% on average (about 1.3% per additional week in the more generous interpretation).
- The wage gain comes from improved match quality — workers find jobs at better-fitting firms — not from higher bargaining power within the same type of match.
- The wage effect is entirely concentrated among workers who find jobs *after* the benefit extension kicks in (late exiters), not those who would have found jobs anyway (early exiters).
- There is a negative cross-study correlation between the unemployment duration effect and the wage effect of UI extensions: in labour markets where UI extends search duration substantially (weak labour markets), the wage gains are smaller, consistent with a duration-dependence force working against the selectivity force.

Mechanism

Two offsetting forces: (i) *selectivity* — workers raise their reservation match quality because the outside option during search is higher, leading to better matches; (ii) *duration dependence* — longer search depletes savings, erodes skills, or signals lower quality to employers, pushing workers into worse matches. The net effect on wages depends on which force dominates. In the Austrian RD, the selectivity effect dominates.

Takeaways for your project

1. **Structural microfoundation.** Your model provides exactly the mechanism Nekoei and Weber (2017) document empirically. In your framework, raising b_U increases the reservation productivity threshold $p^*(x, z)$, making unskilled workers more selective. This is the selectivity channel. Your model can quantify it structurally and decompose it into duration vs. quality components.
2. **Cross-market feedbacks.** Nekoei and Weber (2017) work in a single-market setting. Your model adds the dimension they cannot capture: raising b_U not only improves match quality in the unskilled market, but also shifts the training threshold $\bar{x}$ and changes the composition of the skilled unemployment pool, with further tightness and match-quality effects in the skilled market.
3. **Outside option $\neq$ UI.** The paper uses UI extension as an instrument, but the underlying economics is about outside options. Your Policy Exercise (1) studies outside options generically (b_U, b_S) — the Nekoei and Weber (2017) finding motivates the exercise without requiring you to label b as UI.
4. **The duration–quality trade-off.** Your analytical derivations of expected duration $D_j(x)$ and expected match quality $\mathbb{E}[p \mid p \geq p^*(x, z)]$ as functions of b are precisely the objects needed to replicate the Nekoei and Weber (2017) decomposition structurally.

2 Bárány (2016) — “The Minimum Wage and Inequality: The Effects of Education and Technology”

Main point

The minimum wage affects inequality not only through direct wage compression, but also through its effect on the *education margin*: by changing skill prices, it alters the incentive to acquire education, which changes the skill composition of the workforce, which in turn changes the direction of technical change. The paper builds a GE model that captures all three channels simultaneously.

Specification

Two-sector model (high-skilled, low-skilled) with:

- Heterogeneous workers who differ in ability and time cost of education.
- Endogenous education choice: workers above an ability threshold choose to become high-skilled.
- Endogenous directed technical change: firms direct R&D toward the skill type that is relatively more abundant/profitable, following Acemoglu (2002).
- A binding minimum wage in the low-skilled sector.

- Competitive labour markets (no search frictions).

The education decision is a threshold rule: workers with ability above $\bar{x}$ invest in education. The threshold is determined by the indifference condition where the return to education (higher wage in the skilled sector minus cost) equals zero at the margin.

How she motivates the education margin

- The paper argues that you cannot evaluate minimum wage effects without accounting for the education response. A higher minimum wage compresses the low-skilled wage distribution, which *reduces* the return to education for some marginal workers (the low-skilled sector just got better), but *also* changes the relative supply of skills, which affects directed technical change and thus the skill premium.
- The motivation is that the policy $\rightarrow$ education margin $\rightarrow$ composition $\rightarrow$ GE feedback chain is quantitatively important: ignoring the education response misses a significant part of the effect.
- She estimates the model and finds that the decline in the real minimum wage since the 1980s accounts for a meaningful share of rising wage inequality, once the education and technology channels are accounted for.

Relevance to your project

1. **Same spirit, different frictions.** Bárány (2016)’s model has endogenous education choice with endogenous market composition and GE feedbacks — the same conceptual architecture as your model. The key difference: she has competitive markets, you have search frictions. Your paper can position itself as bringing this education-margin logic into a frictional setting where the continuation values that govern the threshold are fundamentally richer.
2. **Advisor connection.** Citing Bárány (2016) is natural and expected — it is your advisor’s work and it has the closest conceptual structure. The literature review should note the paper’s contribution and distinguish yours by the frictional environment.
3. **Motivation approach.** Observe how she motivates: (a) state the policy question, (b) argue that the education margin matters for that question, (c) show that existing models miss the feedback, (d) build the minimal model that captures it. Your motivation follows the same template but replaces “minimum wage” with “outside options / education subsidies” and “competitive markets” with “frictional markets.”

3 Charlot and Decreuse (2005) — “Self-Selection in Education with Matching Frictions”

Main point

In a frictional two-sector labour market where workers self-select into education, the decentralised equilibrium *always* produces over-education relative to the social optimum. The key mechanism is a composition externality: the marginal entrant into education does not internalise the deterioration she imposes on the mean ability of both the educated and uneducated pools, reducing job creation in both markets.

Model structure

Steady-state, two-sector matching model. Workers are heterogeneous in a single innate ability $a \in [0, 1]$ with density ϕ . Two sectors (high h , low l) each have their own CRS matching function; tightness $\theta_i = v_i/u_i$ is sector-specific. Output in sector i is $y_i(a) = A_i \cdot a$, with $A_h > A_l > 0$. Wages are set by Nash bargaining with bargaining power β ; the resulting wage schedule is linear in ability. Free entry drives the vacancy value to zero, pinning each sector’s tightness as a function of the mean ability of workers in that pool. The two markets are in *simultaneous general equilibrium*: both free-entry conditions and the self-selection condition hold jointly. Job destruction arises only through an exogenous death/replacement rate δ ; there is no endogenous separation. Workers make a binary education choice at birth: pay a fixed, ability-independent cost C (net of any subsidy S) to enter market h , or stay in market l for free.

The equilibrium is characterised by a threshold $\sigma^* \in (0, 1)$ such that workers with $a \geq \sigma^*$ educate. Because both tightness levels depend on the mean ability of each pool, which itself depends on σ^* , the equilibrium is a fixed point in threshold space.

Key findings

1. The unique efficient outcome features a *tax* on education, not a subsidy. Over-education holds for all parametrisations (bargaining power, schooling cost, technology).
2. The composition effect — mean ability of both pools is endogenous to σ^* and feeds back into job creation — is the novel externality specific to frictional markets. It is absent in frictionless competitive settings.
3. A rise in the schooling subsidy lowers σ^* , reducing mean ability in both pools, causing tightness and job-finding rates to fall in both sectors simultaneously.
4. Multiple equilibria are possible when the multiplier (education lowers unskilled pool quality, cheapening the unskilled outside option, encouraging more education) dominates the stabiliser.

Takeaways for your project

1. **Closest structural predecessor.** This paper is the most direct antecedent of your model’s architecture: binary education choice, two markets in simultaneous frictional equilibrium, endogenous composition of both pools, and Nash bargaining throughout. Your model inherits this framework.
2. **Key gap: no job ladder.** Charlot–Decreuse (2005) has no on-the-job search and no job ladder in the skilled market. Wages are determined at matching and never renegotiated due to outside competition. Your model adds a full job ladder (CPR 2006 style) in the skilled market, making the skilled sector’s wage dynamics fundamentally richer and giving it a non-trivial internal wage distribution.
3. **Key gap: no endogenous job destruction.** Separation occurs only through death, ruling out the reservation-quality mechanism that governs your unskilled market (p_U^*). Endogenous job destruction in your unskilled market is what makes the outside-option exercise (b_U) tractable: higher b_U shifts p_U^* , not merely search intensity.
4. **Key gap: ability-independent cost.** The cost C is the same for all workers. Your training cost $c(x) = c_0(1-x)e^{-x}$ declines in ability, making the composition externality richer and the policy analysis of cost subsidies more nuanced.
5. **Key gap: no aggregate states.** Your model adds transitions between aggregate states $z \in \mathcal{Z}$, so the training threshold $\bar{x}(z)$ is endogenous to the business cycle — a dimension entirely absent here.
6. **Correct the literature table.** The existing table states the gap as “composition exogenous, markets not jointly in equilibrium,” which is incorrect for this paper. Both the composition *and* the simultaneous equilibrium are present here. The genuine gap is: no job ladder, no aggregate states, and an ability-independent schooling cost.

4 Charlot et al. (2005) — “Adaptability, Productivity, and Educational Incentives in a Matching Model”

Main point

When education raises both a worker’s productivity within a job and her *adaptability* (the fraction of technologies she can operate), the direction of the education-market inefficiency depends on which dimension dominates. Productivity-only education is under-supplied; adaptability-only education is over-supplied. In the general case, either can hold.

Model structure

Continuous time, a continuum of technologies $j \in [0, 1]$ each with its own sub-market but with identical tightness θ in symmetric equilibrium. Workers are *homogeneous in ability* and choose a continuous schooling *duration* T_i rather than a binary type. Education raises:

(i) *productivity* $y(T_i)$ on any matched job, and (ii) *adaptability* $F(T_i)$, the share of technologies the worker can operate, so her contact rate is $F(T_i) \cdot m(\theta)$. Education is purely time-consuming (opportunity cost only; no tuition). All workers choose the same T^* in equilibrium. Wages are Nash bargaining with parameter β . Free entry pins market tightness θ^* . The equilibrium is a pair (T^*, θ^*) satisfying simultaneously the Optimal Schooling condition and the Free-Entry condition. A unique equilibrium exists.

Key findings

1. When education raises only productivity: under-investment, because higher productivity raises vacancy creation (a positive externality workers do not internalise).
2. When education raises only adaptability: over-investment, because the extensive wage return to adaptability (improving one's outside option in bargaining) is privately valuable but not socially so, and congestion in the additional sub-markets harms others.
3. Both effects combined admit either sign for the inefficiency.
4. Higher job destruction q raises schooling duration when adaptability dominates — consistent with European data from the 1970s–1980s showing simultaneous rises in education and unemployment.

Takeaways for your project

1. **No pool segmentation.** All workers are identical; there is a single representative-agent optimum, not a threshold splitting a heterogeneous population into two markets. Your model's composition externality (endogenous mean ability of two pools feeding back into job creation in both) is absent here.
2. **Continuous vs. binary training.** Your model features a binary threshold $\bar{x}(z)$, closer to Charlot–Decreuse (2005). The continuous duration choice here may be relevant if you introduce a schooling-length margin in future extensions, but is not part of your current structure.
3. **No two-market simultaneous equilibrium.** While the market is segmented by technology, all workers compete in a single aggregate market with a single tightness. The two-market structure with separate equilibria is your paper's key contribution.
4. **No job ladder.** No PVR sequential auction, no on-the-job search, no wage dynamics within a sector.
5. **No aggregate states.**

5 Wasmer (2006) — “General versus Specific Skills in Labor Markets with Search Frictions and Firing Costs”

Main point

In a single frictional labour market with firing costs, the equilibrium skill regime (general or specific human capital) is determined by the level of search frictions: high frictions favour specific skills (which raise within-firm productivity at the cost of portability), while low frictions favour general skills (which preserve outside options at the cost of within-firm productivity). Employment protection (firing taxes) pushes economies toward specific-skill equilibria, rationalising transatlantic differences in turnover and training.

Model structure

Standard Mortensen–Pissarides (1994) extension to a *single* labour market. Workers are *homogeneous in ability*; each newborn chooses between specific (s) or general (g) human capital at job entry. Production: $y = \varepsilon + h^k$, where ε is an idiosyncratic match-specific productivity shock (Poisson arrival rate λ , drawn from distribution F) and h^k is the worker’s human capital level ($h^s > h^g$ within firm; general capital is retained across jobs, specific capital is lost on separation). Wages are Nash bargaining at two stages: an entry wage and a renegotiated continuation wage after shocks. Job destruction is endogenous: a match is destroyed when surplus $S_k(\varepsilon) < 0$, defining a reservation productivity R^k . Free entry pins tightness θ^K . Three regimes are possible: S (all specific), G (all general), and M (mixed). A firing tax T (paid by the firm, non-transferable to worker) is included.

Key findings

1. Workers prefer specific skills when job-finding rate p is low and general skills when p is high. The threshold p^M separates the two pure regimes.
2. Specific skills and low turnover reinforce each other (complementarity): the S-regime features longer job duration and higher within-firm productivity.
3. Higher employment protection (firing tax T) raises the likelihood of the S-regime by reducing tightness θ and hence lowering p .
4. Application: continental Europe’s higher firing costs $\rightarrow$ specific skills $\rightarrow$ lower turnover $\rightarrow$ higher long-term unemployment.

Takeaways for your project

1. **Single market, not two markets.** This is the key distinction from your paper. Wasmer (2006) has one labour market in which all workers — regardless of skill type

— compete. There is no separation into a skilled and an unskilled market with different tightness levels, different wage distributions, and separate free-entry conditions. The existing literature table’s description (“two markets not simultaneously in equilibrium”) is misleading: the correct characterisation is that this paper has *no market segmentation at all*.

2. **No composition externality.** The fraction κ^g in the mixed regime is determined by a representative-agent indifference condition, not by a threshold over a heterogeneous distribution. There are no pools whose mean characteristics feed back into job creation.
3. **No job ladder.** Workers stay in their current firm until idiosyncratic shocks trigger separation. There is no PVR sequential auction, no on-the-job wage dynamics, no job-to-job mobility. Wage renegotiations respond only to idiosyncratic ε shocks, not to outside offers.
4. **On-the-job vs. pre-market training.** The baseline model has workers choosing skills at job entry. Although Section IVB shows the schooling extension produces identical steady states, the fundamental structure remains a single-market model.
5. **Endogenous job destruction.** Wasmer (2006) is a useful reference for the endogenous job-destruction mechanism in your *unskilled* market, where the DMP reservation-quality rule p_U^* plays an analogous role to R^k here.
6. **No aggregate state transitions.**

6 Charlot and Decreuse (2010) — “Over-Education for the Rich, Under-Education for the Poor”

Main point

When workers are heterogeneous in *both* ability and schooling cost, the decentralised frictional equilibrium simultaneously over-educates low-ability, low-cost (“rich”) workers and under-educates high-ability, high-cost (“poor”) workers. The mechanism is a market adverse selection: rich workers crowd the educated pool, reducing its mean ability and depressing job creation in the high-skill sector, which undermines the return to schooling for the talented poor.

Model structure

Direct extension of Charlot and Decreuse (2005) with *two-dimensional worker heterogeneity*: ability a and schooling cost c , jointly distributed with CDF Φ on $\mathbb{R}_+^2$. Two sectors (h, l) with $y_{ia} = A_i \cdot a$, $A_h > A_l$; each has a CRS matching technology with tightness θ_i . Search in each sector is random (ability is unknown to firms ex ante), so the free-entry condition in sector i depends on mean ability $\bar{a}_i$ of the job-seekers in that pool. Wages are Nash bargaining; job destruction is exogenous (death rate δ). The two markets are in simultaneous general equilibrium. The education decision: worker (a, c) chooses education if $U_{ha} - c \geq \max\{U_{la}, 0\}$.

This defines a linear indifference locus $c = R^* \cdot a$ in the (a, c) plane, where the endogenous return R^* depends on tightness in both sectors. The equilibrium is a fixed point in mean abilities $(\bar{a}_h, \bar{a}_l)$.

Key findings

1. Simultaneous over-education for the rich (a low, c low) and under-education for the poor (a high, c high), provided ability and schooling cost are not too positively correlated.
2. The private return R^* is independent of ability (a flat indifference line in (a, c) space), while the social return $R^s(a)$ is increasing in a . Hence the rich over-invest and the poor under-invest.
3. Optimal policy is redistributive: a subsidy proportional to schooling cost, combined with a lump-sum tuition fee. Net, education should be taxed, not subsidised.
4. Generalises Charlot and Decreuse (2005): with degenerate cost distribution, the model collapses to universal over-education.

Takeaways for your project

1. **Two-dimensional heterogeneity.** Your model features ability heterogeneity (type x) and an ability-dependent training cost $c(x) = c_0(1 - x)e^{-x}$. This creates a single-dimensional threshold $\bar{x}(z)$: unlike the 2010 paper's linear locus in (a, c) space, your threshold is a scalar. The key distinction is that your cost is a *known function* of type, not an independent draw, so there is no two-dimensional selection problem.
2. **No job ladder in the skilled market.** As in Charlot–Decreuse (2005), the high-skill sector here operates with static Nash bargaining and no PVR wage dynamics. Your model's job ladder is the central innovation not present in this paper.
3. **Simultaneous two-market equilibrium.** Your paper shares this with Charlot–Decreuse (2010). The key addition is the job ladder and aggregate states.
4. **Policy exercise connection.** The 2010 paper's analysis of redistributive education subsidies — differentiating between ability and cost dimensions — is directly relevant to your Policy Exercise (2): comparing flow training subsidies (reducing b_T) vs. upfront cost reductions (reducing c_0). Your framework allows you to quantify both effects in a calibrated model.
5. **No aggregate states.** Charlot–Decreuse (2010) is a steady-state model. Your model's aggregate states are the source of the cyclical training threshold, absent here.

7 Charlot and Malherbet (2013) — “Education and Employment Protection”

Main point

The paper studies the link between employment protection legislation (EPL) and educational investment in a Mortensen–Pissarides matching model with dual labour contracts (temporary and permanent) and endogenous schooling. The key result is that firing costs and temporary employment exert *opposite* effects on the education margin: firing costs stabilise employment relationships, raising the return to human capital and spurring education; but the expansion of short-term contracts increases labour turnover, reducing the return to education. Laissez-faire produces insufficient educational investment (a hold-up problem under Nash bargaining), and while firing costs can be welfare-improving by raising education incentives, dual EPL — the European model of high firing costs combined with a large temporary-contract sector — largely cancels out any educational gain.

Model structure

Continuous-time steady-state Mortensen–Pissarides (1994, 1999) matching model with the following extensions:

- **Endogenous education.** Workers choose education level $e \geq 0$ at the start of working life, at a linear cost $c(e) = c_0 e$. Education raises match-specific productivity $y(e)$ on all jobs, with $y' > 0$, $y'' < 0$. Education is general and is completed before entering the labour market.
- **Match-specific productivity shocks.** A match starts at productivity x_0 ; idiosyncratic shocks arrive at Poisson rate λ drawn from distribution G on $[x_l, x_u]$. A match survives if $x \geq x_R$, the endogenous reservation productivity.
- **Dual EPL.** Two contract types: (i) Long-term jobs (LTJ) protected by a firing cost f (red tape, non-transferable) applied at renegotiation; (ii) Short-term jobs (STJ) terminated at no firing cost with probability ν per unit time, convertible to LTJ when profitable. Fraction p of new hires on STJ (regulated by law). Wages via Nash bargaining in both regimes.
- **Education decision.** Worker chooses e to maximise lifetime utility, with the first-order condition equating $y'(e)$ to the frictional private return. Because Nash bargaining extracts share β of the surplus, the private return is a fraction of the social return — the source of the hold-up and underinvestment.

Key findings

1. Laissez-faire produces insufficient educational investment: Nash bargaining creates a hold-up in pre-market education, exactly analogous to Acemoglu and Shimer (1999).

2. Firing costs alone (without dual EPL) can be welfare-improving: by stabilising long-term employment, they raise the rate of human capital utilisation and increase the private return to education.
3. The first-best policy is a direct education subsidy, not institutional mandates on job protection.
4. In a dual EPL context, the education-reducing effect of STJ turnover nearly cancels or dominates the education-enhancing effect of firing costs. Dual EPL generates small or zero net welfare gains.
5. The analysis is consistent with Blanchard–Landier (2002) and Cahuc–Postel-Vinay (2002): dual labour markets are welfare-inferior to a uniform contract.

Takeaways for your project

1. **Hold-up in education + Nash bargaining.** Charlot–Malherbet identify exactly the same hold-up mechanism as Acemoglu and Shimer (1999): workers invest in education before matching, Nash bargaining means they receive only fraction β of the return, causing underinvestment. In your model, the training threshold $\bar{x}(z)$ in the decentralised equilibrium is too high relative to the social optimum through this channel — flow training subsidies that reduce the effective cost improve welfare by correcting this distortion.
2. **Endogenous job destruction coupled with training.** Charlot–Malherbet couple endogenous job destruction (via the reservation productivity x_R) with the education decision. In your model, endogenous separation governs both markets (the DMP reservation rule p_U^* in the unskilled market; the OJS cutoff p_S^{oj} in the skilled market). Anything that changes job stability — higher b_U reducing p_U^* — alters the return to training, a channel Charlot–Malherbet formalise.
3. **Direct subsidy as the first-best.** Their finding that education subsidies dominate EPL interventions corroborates your policy exercise (2): a flow training subsidy that directly targets the training decision is a first-best instrument, while changes in market conditions (b_U , b_S) are at best second-best.
4. **Single market, no job ladder, no aggregate states.** The key gap from your paper: Charlot–Malherbet have one undifferentiated labour market (all workers in the same frictional pool regardless of education level), no market segmentation, no job ladder in the skilled market, and no aggregate state transitions. Your paper adds: (i) two separately clearing markets with different tightness levels, (ii) CPR job ladder in market S , and (iii) the cyclical training threshold $\bar{x}(z)$.
5. **Literature table correction.** The previous entry for this paper (“file mismatch note”) should be replaced. The paper is a Family 1 predecessor: it combines frictional matching with endogenous education, features a hold-up problem under Nash bargaining, and derives welfare implications of education subsidies — but within a single-market framework, without a job ladder or aggregate states.

8 Diamond (1982) — “Wage Determination and Efficiency in Search Equilibrium”

Main point

Wage bargaining and search frictions jointly determine equilibrium unemployment. Two-sided search externalities prevent the decentralised outcome from replicating the social optimum, even when technology is homogeneous.

Model structure

Steady-state partial equilibrium with fixed-coefficient technology: each filled job produces constant output y . Matches form through a stochastic search process; breaks occur at exogenous Poisson rate b . Wages are set by symmetric Nash bargaining, splitting the bilateral surplus equally. Asset-value equations for unemployment W_U , employment W_E , vacant jobs W_V , and filled jobs W_F close the model. The equilibrium wage is:

$$\frac{w}{y} = \frac{rb^{-1} + u^{-1}}{2rb^{-1} + u^{-1} + v^{-1}}.$$

Search externalities arise because an additional unemployed worker speeds vacancy filling but slows job finding for other workers.

Takeaways for your project

Diamond (1982) supplies the core conceptual architecture: the bilateral meeting process, asset-value (Bellman) representation of worker and firm states, Nash bargaining wage determination, and the insight that equilibrium wages depend on market tightness (u, v) . Your model inherits all of these. Diamond is the building block on which Pissarides (2000) and all subsequent search foundations in this reading list rest.

9 Pissarides (2000) — *Equilibrium Unemployment Theory*

Main point

The canonical textbook treatment of the DMP model, covering: the matching function, free entry, Nash bargaining, endogenous job destruction, and efficiency. Provides the complete structural skeleton for search-based macro-labour models.

Key chapters relevant to your model

Chapter 1 (DMP core). Matching function $m(u, v)$ with CRS; tightness $\theta = v/u$; vacancy-filling rate $q(\theta)$; job-finding rate $\theta q(\theta)$. Free-entry job-creation condition: $c/q(\theta) = J^F$. Nash bargaining wages. Beveridge curve: $\delta(1 - u) = \theta q(\theta)u$. The Hosios (1990)

efficiency condition: $\beta = \eta$ (worker bargaining power equals matching-function unemployment elasticity).

Chapter 2 (Endogenous job destruction). Match-specific productivity drawn from a distribution and shocked over time. Matches are destroyed when productivity falls below a *reservation productivity* R , determined by $S(R) = 0$. Simultaneous determination of job creation and job destruction. This chapter directly founds your unskilled market’s endogenous separation rule p_U^* .

Chapter 4 (On-the-job search). Workers search on the job and move directly to better matches. Directly relevant to your skilled market.

Chapter 8 (Efficiency). With endogenous job destruction, efficiency requires the Hosios condition for creation *and* correct internalisation of the option value of search and the cost of destruction.

Takeaways for your project

Pissarides (2000) provides the complete structural skeleton of your model: the matching function, free-entry vacancy posting, the Beveridge curve, Nash bargaining, and (from Ch. 2) the endogenous job-destruction mechanism. Market tightness θ as the key state variable linking job-finding and vacancy-filling rates is adopted throughout your framework.

10 Postel-Vinay and Robin (2002) — “Equilibrium Wage Dispersion with Worker and Employer Heterogeneity”

Main point

A two-sided heterogeneous search model with on-the-job search in which wages are set through a *sequential auction* (Bertrand competition among employers) generates an endogenous, dispersed cross-sectional wage distribution. This breaks the wage-compression result of Burdett–Mortensen and provides a structural foundation for the empirical wage distribution.

Model structure

Continuous-time, steady-state. Workers differ in ability ε (time-invariant); firms differ in productivity p ; match output is εp . Unemployed workers receive offers at rate λ_0 ; employed workers at rate λ_1 . When an employed worker receives an outside offer, a *poaching competition* (Bertrand auction) begins: the less productive employer reaches its willingness-to-pay ceiling first and the worker either stays at a higher wage or moves. This generates a *two-wage structure* for each worker type: a low entry wage (hired from unemployment) and a high “promotion” wage (triggered by outside competition). Estimated on French matched employer–employee data; frictions explain roughly 50% of wage dispersion.

Takeaways for your project

PVR is the direct ancestor of your **skilled market** wage-setting protocol. You inherit: (i) the sequential auction mechanism, (ii) the two-wage structure (entry wage when hired from unemployment; renegotiated wage after outside offer), (iii) the framework for identifying firm and worker heterogeneity distributions, and (iv) the endogenous wage dispersion driven by the history of outside offers. Your model extends the static PVR setting by adding aggregate shocks and a second (unskilled) market with a separate, simpler Nash bargaining protocol.

Important distinction: your model uses **Nash bargaining** (CPR 2006 style) in *both* markets, not the full PVR sequential auction. In PVR, firms have all bargaining power over unemployed hires. Your skilled market combines the CPR game-theoretic microfoundation (positive worker bargaining power $\beta > 0$) with the on-the-job Bertrand competition.

11 Shimer (2005) — “The Cyclical Behavior of Equilibrium Unemployment and Vacancies”

Main point

The standard DMP model with Nash wage bargaining cannot amplify plausible productivity shocks enough to generate observed unemployment and vacancy volatility (the “Shimer puzzle”). The culprit is fully flexible Nash wages: positive productivity shocks raise workers’ outside option, absorbing the gain into wages and leaving vacancy creation nearly unchanged.

Model structure and findings

Canonical MP model with two aggregate stochastic shocks (labour productivity p and separation rate). Calibrated to U.S. data 1951–2003. Key data facts: the standard deviation of the v – u ratio is approximately 20 times that of labour productivity, while the model predicts near-equal volatility. Unemployment and vacancies are strongly negatively correlated (-0.89). The job-finding rate co-moves tightly with the v – u ratio. The Beveridge curve is the downward-sloping locus of (u, v) pairs consistent with stock-flow balance.

Takeaways for your project

Shimer (2005) is the benchmark quantitative problem your model is designed to improve upon. You inherit: (i) the calibration methodology for search models, (ii) the set of business-cycle moments your model must match (volatility of unemployment, vacancies, v – u ratio, job-finding rate), (iii) the diagnostic that Nash bargaining causes wages to absorb too much productivity variation, and (iv) the Beveridge curve as an empirical target. Later papers (Robin 2011; Lise and Robin 2017; Moscarini and Postel-Vinay 2024) all take Shimer’s puzzle as their point of departure.

12 Cahuc et al. (2006) — “Wage Bargaining with On-the-Job Search: Theory and Evidence”

Main point

The PVR sequential auction can be given a rigorous game-theoretic foundation via three-player Rubinstein alternating-offers bargaining. This embeds positive worker bargaining power $\beta > 0$ in the wage-setting protocol, making the model empirically estimable with firm-level productivity data.

Model structure

Continuous time. Workers heterogeneous in ability ε ; firms heterogeneous in productivity p ; match output εp . Unemployed workers receive εb (proportional to ability). Layoff rate δ exogenous. Key innovation: when an employed worker receives an outside offer, a *three-player Rubinstein alternating-offers game* begins among the worker, current employer, and poaching employer. Under continuous time and equal discount rates, this game has a unique subgame-perfect equilibrium that replicates the PVR sequential auction when $\beta \rightarrow 0$ and approaches Nash bargaining when the outside offer is close to the current match value. Estimated on 1993–2000 French matched employer–employee administrative panel; between-firm competition explains more wage gains than Nash bargaining power, especially for unskilled workers (estimated $\beta \approx 0$).

Takeaways for your project

CPR (2006) is the **direct structural foundation** for your skilled-market wage equation. You inherit: (i) the theoretical justification for using the sequential auction when workers have positive β , (ii) the identification strategy exploiting both wage and productivity data, (iii) the result that competition matters quantitatively more than bargaining for low-skill workers (relevant to your unskilled market, where you use standard Nash bargaining), and (iv) the continuous-time steady-state structure with heterogeneous workers and firms. The CPR wage equation (or your discrete-time analog) is a core building block in your skilled market. Your unskilled market, by contrast, uses the simpler Nash bargaining without the sequential auction.

13 Robin (2011) — “On the Dynamics of Unemployment and Wage Distributions”

Main point

Embedding PVR-style sequential auctions in a dynamic stochastic setting with aggregate productivity shocks and endogenous job destruction resolves the Shimer puzzle through a *heterogeneity amplification* mechanism: negative shocks push workers at the margin of viable

matches into unemployment, generating large, realistic fluctuations from modest productivity volatility.

Model structure

Discrete time; aggregate state $y_t \in \{y_1 < \dots < y_N\}$ (finite Markov chain). M worker types with ability x_m ; firms are *identical* (no firm heterogeneity). Match output: $y_i(m) = x_m y_i$. Both exogenous (δ) and endogenous destruction (when surplus turns negative) operate simultaneously. PVR wage contracting: starting wages at reservation value, renegotiated only when outside offers arrive. Long-term wage contracts create wage stickiness without exogenous rigidity. If 6% of workers are near the viability margin, unemployment can swing from 4% to 10% with only modest productivity volatility. Estimated by simulated GMM. The tails of the wage distribution are more procyclical than middle wages.

Takeaways for your project

Robin (2011) is your most direct antecedent in the PVR line. You inherit: (i) the discrete-time, aggregate-shock extension of PVR that founds your aggregate-state Markov chain $z \in \mathcal{Z}$, (ii) endogenous job destruction as an amplification mechanism (your skilled market's on-the-job search cutoff p_S^{oj} plays an analogous role), (iii) the two-wage structure under sequential auctions with aggregate shocks, (iv) the recursive DSGE formulation with a single aggregate state variable, and (v) the simulated GMM estimation strategy on U.S. data. Your paper extends Robin (2011) in three dimensions: adding a second (unskilled) market with a separate equilibrium, endogenising the composition of both labour pools via the training threshold $\bar{x}(z)$, and using Nash bargaining throughout (not the pure PVR monopsony-at-entry protocol).

14 Bagger et al. (2014) — “Tenure, Experience, Human Capital, and Wages: A Tractable Equilibrium Search Model of Wage Dynamics”

Main point

A tractable equilibrium job search model that jointly estimates human capital accumulation (driven by experience) and employer heterogeneity within the PVR/CPR sequential auction framework. The key tractability device is the piece-rate contract; the key empirical finding is that job shopping (between-job mobility early in the career) is quantitatively more important than human capital accumulation for early wage growth.

Model structure

Three combined elements: (i) human capital accumulation (general, driven by experience), (ii) employer heterogeneity (firms differ in productivity p), and (iii) idiosyncratic productivity

shocks. Piece-rate contracts ($w = \phi \cdot \varepsilon_t \cdot p$, piece rate ϕ renegotiated via sequential auction upon outside offers) achieve tractability under all three. Estimated on matched employer–employee data using both firm output and worker wages. Wage growth decomposed into: on-the-job human capital accumulation, within-job wage increases from Bertrand competition, and between-job wage increases from job shopping. The “job-shopping phase” (first ~ 10 years) is dominated by between-job mobility.

Takeaways for your project

If your model features experience-wage profiles or within-market human capital, Bagger et al. is the reference. More broadly, you inherit: (i) the piece-rate device for tractability under idiosyncratic shocks, (ii) the decomposition of wage growth into human capital and search components, (iii) the insight that firm heterogeneity and worker heterogeneity together are needed to separately identify rent-sharing from human capital, and (iv) the estimation methodology using matched data on both output and wages. In your current model, training changes market access but not within-market human capital, so Bagger et al. is more of a methodological benchmark than a direct structural predecessor.

15 Lise et al. (2016) — “Matching, Sorting and Wages”

Main point

An empirical search-matching model with two-sided heterogeneity, productive complementarities, endogenous job destruction, and on-the-job search shows that the AKM firm–worker correlation is a misleading measure of sorting: structural estimation is needed to correctly identify the degree of positive assortative matching. Welfare costs of mismatch are near zero for unskilled workers and substantial for college graduates.

Model structure

Draws on Mortensen–Pissarides (1994) for endogenous match creation and destruction (separation when idiosyncratic shock drives surplus to zero) and on PVR (2002)/CPR (2006) for wages (Nash bargaining for new matches from unemployment; Bertrand competition for poaching). Key innovation: *productive complementarities* between worker ability x and job type p via a non-separable output function $f(x, p)$, enabling positive assortative matching for output reasons. Estimated on NLSY with simulated method of moments. Welfare decomposition: unskilled labour market near-zero complementarity; college-educated market significant positive complementarity.

Takeaways for your project

LMR is relevant if your paper deals with sorting or job quality differentials across the two markets. You draw on their framework for: (i) combining endogenous job destruction (MP) with the sequential auction (PVR/CPR) and on-the-job search, (ii) the concept of match-specific surplus as the object governing all mobility decisions (block-recursivity), (iii) welfare

decomposition methodology, and (iv) the treatment of worker–firm complementarities as an identification target. Your model is simpler in that it treats match quality as drawn from a distribution $\Gamma(\cdot \mid z)$ rather than modelling the full (x, p) complementarity function, and adds the education-choice margin that LMR omit.

16 Lise and Robin (2017) — “The Macro-dynamics of Sorting between Workers and Firms”

Main point

Proving block-recursivity in a two-sided heterogeneous search model with aggregate shocks makes tractable the first empirically estimated model of cyclical sorting between workers and firms. Aggregate shocks affect two margins: during recessions, low-type workers matched with high-type firms are fired; during recoveries, low-type workers are re-absorbed.

Model structure

Extends PVR (2002) with aggregate TFP shocks and extends Robin (2011) with firm heterogeneity and Pissarides-style free-entry vacancy posting. Match output is a function of the worker–firm type pair, allowing complementarities. Firms post state-contingent offers under the sequential auction. Key tractability result: the *match surplus function* does not depend on the current wage, on the distribution of unemployed workers, on the distribution of vacancies, or on the employment distribution — **block-recursivity**. The fixed point defining the surplus is solved independently of endogenous distributions. Fitted to aggregate U.S. labour market data 1951–2012. The matching set is cone-shaped (strong positive sorting).

Takeaways for your project

Lise and Robin (2017) is a key technical precedent. You inherit: (i) the block-recursivity result and the technique of working with the joint surplus rather than separate value functions, (ii) the method of characterising the matching set (which worker types accept which firm types) under two-sided heterogeneity and aggregate shocks, (iii) vacancy creation via free entry grafted onto the PVR wage protocol, (iv) cyclical sorting dynamics as an empirical phenomenon your model should account for, and (v) the estimation approach combining aggregate time-series data with cross-sectional dispersion moments. Your model takes worker-type distribution as exogenous within each market *conditional* on the training threshold, where Lise and Robin take the full distribution as given. Your paper endogenises the threshold itself.

17 Acemoglu (2001) — “Good Jobs versus Bad Jobs”

Main point

In a two-type DMP framework (good vs. bad jobs differing in sunk creation costs), the decentralised equilibrium produces too few good jobs relative to the social optimum, because good-job firms generate a larger positive externality through rent-sharing that they do not internalise (hold-up problem). Minimum wages and unemployment benefits can improve welfare by correcting job composition.

Model structure

Extension of the standard DMP model with two job types (g and b). Good jobs require larger sunk creation cost $k_g > k_b$; both types sell into competitive markets with imperfect substitution between outputs ($p_g > p_b$). Workers are homogeneous. Search is undirected; matching function $M(u, v)$ with CRS. Wages: asymmetric Nash bargaining with worker power β . Free entry drives vacancy values to their respective creation costs. The composition of jobs is determined by the relative profitability of posting each type. A hold-up problem arises: good-job firms sink large costs before bargaining, share more rent with workers, and create a larger positive externality that they do not internalise. Multiple equilibria are possible (many good jobs + high outside options; vs. many bad jobs + low outside options).

Takeaways for your project

Acemoglu (2001) is directly relevant to the job-quality dimension of your two-market framework. You inherit: (i) the two-type job framework within a DMP search environment — your unskilled market (“bad jobs”) and skilled market (“good jobs”) map directly onto this structure, (ii) the hold-up logic linking sunk creation costs to equilibrium composition distortions, (iii) the analysis of how labour market policies affect not just the level but the *composition* of employment, (iv) the general equilibrium feedback through the worker’s outside option, and (v) the possibility of multiple equilibria. Acemoglu’s externality is distinct from Charlot–Decreuse’s composition externality but operates through a related channel: in both cases, the marginal entrant into the better market imposes a cost on others that she does not internalise.

18 Moscarini and Postel-Vinay (2024) — “On the Job Search and Business Cycles”

Main point

Embedding on-the-job search (OJS) with PVR sequential auctions in a DMP business-cycle framework generates three amplification and propagation channels that the standard model entirely lacks: a matching-function elasticity channel, a search-pool composition channel,

and an employment-distribution channel. The distribution of employment across the job ladder is an endogenous, slowly-evolving state variable that propagates aggregate shocks.

Model structure

DMP-style with OJS; both employed and unemployed workers search with different intensities. Firms are ex post heterogeneous in match productivity (drawn upon meeting), generating a job ladder. Free-entry vacancy posting. Wages: PVR sequential auction (Bertrand competition, rank-preserving equilibrium). Because the equilibrium is rank-preserving, the distribution of employment across the job ladder follows a tractable closed-form Markov process. Three OJS channels identified: (1) *matching-function elasticity* — accounting for OJS raises estimated vacancy elasticity from 0.32 to 0.50; (2) *search pool composition* — employed searchers (more expensive to hire) are more prevalent in booms, dampening job creation; (3) *employment composition / job ladder* — after a negative TFP shock, reallocation slows, employment deteriorates (workers clustered in lower-quality matches), gradually stimulating job creation via endogenous persistence.

Takeaways for your project

MPV (2024) is peripheral but relevant as a high-level orientation. Your model also features OJS in the skilled market and a job ladder generating cyclical employment-distribution dynamics. You inherit from MPV: (i) the three amplification channels as an organising framework for interpreting your model’s cyclical predictions, (ii) the tractable DMP framework with OJS and free-entry vacancy posting under the sequential auction, (iii) the rank-preserving equilibrium result, which simplifies computation of the employment distribution, and (iv) the empirical targets from Shimer (2005) as moments to discipline calibration. If your paper includes a section on business-cycle amplification, MPV provides the direct benchmark. Mark as peripheral in the literature tables.

19 Heckman et al. (1998) — “Explaining Rising Wage Inequality”

Main point

Heckman et al. (1998) build a dynamic overlapping-generations general equilibrium model with heterogeneous human capital to explain rising U.S. wage inequality since the 1960s. The central insight is that wages diverge from skill prices because of endogenous on-the-job investment, and that skill-biased technical change — rather than immigration or trade — is the quantitatively dominant driver of observed inequality trends. The general equilibrium supply response to higher skill prices gradually compresses the college premium, generating the observed non-monotone path of wage differentials over 35 years.

Model structure

Overlapping generations, general equilibrium, following Ben Porath (1967) and Auerbach–Kotlikoff (1987). Agents are heterogeneous in two dimensions: (i) initial ability θ (permanent); (ii) cohort position in the OLG structure. Each worker makes a discrete schooling choice $S \in \{\text{low, high}, \dots\}$ at the start of working life, which maps into access to different on-the-job training technologies.

- **CES aggregate production:** $Y = F(K, H_S, H_U)$ with imperfect substitution across skill groups. Within-group perfect substitution: aggregate supply is $H_S = \int h_S(\theta, a) d\mu(\theta, a)$.
- **On-the-job training (OJT):** After schooling, each worker solves a Ben Porath problem, choosing the fraction of time devoted to producing new human capital: $\dot{h} = g(h, I) - \delta h$. OJT technologies differ by schooling level, so wage profiles diverge within schooling groups.
- **Skill-biased technical change (SBTC):** $F(K, H_S, H_U)$ includes a time-varying factor A_t shifting the relative marginal product of H_S . Wages $w(\theta, a, t) = p_S(t) \cdot h(\theta, a)$ diverge from skill prices during OJT investment.
- **Estimation:** Model parameters estimated by matching micro-predictions on human capital accumulation to NLSY/PSID data; macro technology parameters pinned by aggregate data.

Key findings

1. A model with estimated SBTC can quantitatively match both the 1979–1987 rise in the college–high school wage differential and the 1960–1979 stability and re-rise.
2. Immigration of unskilled workers contributes negligibly; the required magnitudes would be implausibly large.
3. Skilled workers initially exhibit *falling* measured wages at the onset of SBTC: they invest more on the job in anticipation of future high prices, depressing current wages — a pure investment timing effect.
4. General equilibrium supply responses dampen but do not eliminate long-run skill-premium increases.
5. Inequality *within* education groups is generated by ability differences interacting with differential OJT responses.

Takeaways for your project

1. **GE supply-response discipline.** HLT’s central message is that micro evidence on education supply responses must be embedded in a GE model before policy exercises are informative. Your model must account for GE changes in market tightness and skill-sector wages when varying training subsidies — analogous to HLT’s skill-price GE feedback.

2. **Skill prices vs. wages.** In your model, wages are determined by the CPR sequential auction. The wage observed by the econometrician during SMM estimation reflects both search rents and human capital, mirroring HLT’s distinction between observed wages and underlying skill prices. SMM moments should account for this composition.
3. **Training cost heterogeneity and ability.** HLT treats heterogeneity in initial ability θ as the key selection device. Your training threshold $\bar{x}(z)$ depends on aggregate state z ; HLT justifies having heterogeneous agents sorted by an ability-related cost.
4. **No search frictions.** HLT has a competitive frictionless labour market — wages equal skill prices up to an investment wedge. The entire motivation for your model is to add frictional markets, so HLT serves as the competitive benchmark that your model approaches only in the limit of zero frictions.

20 Abbott et al. (2019) — “Education Policy and Intergenerational Transfers in Equilibrium”

Main point

Abbott et al. (2019) build a quantitative OLG life-cycle model with incomplete markets, intergenerational human capital transmission, and heterogeneous agents to evaluate the equilibrium effects of college financial aid policies. The core finding is that crowding out of parental transfers by government aid is large and cannot be ignored in policy analysis: roughly 50 cents per dollar of additional government aid is offset by reduced parental contributions. The current U.S. federal aid system raises long-run welfare by 6% relative to a no-aid counterfactual; ability-tested grants dominate both expanded loans and tax cuts.

Model structure

Overlapping generations life-cycle model:

- **Agents.** A continuum of parent–child pairs. Children differ in cognitive ability α_c and non-cognitive ability κ_c , transmitted stochastically from parents: $(\alpha_c, \kappa_c) = \Phi(\alpha_p, \kappa_p, \varepsilon)$.
- **Education decision.** A young adult decides whether to attend college (binary). The psychic cost of college depends on (α_c, κ_c) . College can be funded by parental transfers, government grants, government loans, private loans, or student labour during college.
- **Labour market.** Workers of different gender and education are imperfect substitutes in a CES production function; wages are competitively determined skill prices with no search frictions.
- **Incomplete markets.** Workers face idiosyncratic earnings risk; no perfect insurance.

- **Crowding out.** When government aid rises, rational parents reduce intergenerational transfers by nearly the same amount (for constrained parents), leaving net resources roughly constant.
- **Policy experiments.** Changes in grant generosity, loan limits, and income tax cuts evaluated in GE transition and long run.

Key findings

1. Crowding out of parental transfers is roughly 50 cents per dollar of additional government aid.
2. The current U.S. federal aid system raises long-run welfare by 6% relative to no-aid.
3. Ability-tested grants dominate loan expansion and tax cuts in welfare terms.
4. GE feedbacks are important: expanded college enrollment reduces the college premium, partially offsetting private returns.

Takeaways for your project

1. **Crowding out and policy impact.** Raising b_U (outside options) may have analogous crowding-out effects on private training investment: if workers' outside options rise, the marginal return to training falls, potentially reducing training rates. AGMV shows that ignoring behavioural substitution overstates policy impact.
2. **Upfront cost reduction vs. flow subsidies.** AGMV's finding that ability-tested grants are more effective than loan expansion parallels your Policy Exercise (2): upfront cost reductions (like grants) vs. flow training subsidies. Upfront cost reductions are likely more distortion-free because they target the entry margin directly.
3. **GE price effects.** The college premium falls in AGMV when enrolment expands. In your model, the analogous effect is that market tightness in the skilled sector θ_S adjusts when more workers train, changing wages through the CPR mechanism. SMM estimation must incorporate the GE tightness response.
4. **No frictions.** Like Heckman et al. (1998), AGMV has no search frictions; wages equal marginal products. Your model is the frictional version. The comparison with AGMV benchmarks how much of training behaviour is explained by frictions vs. financial constraints.

21 Acemoglu and Shimer (1999) — “Holdups and Efficiency with Search Frictions”

Main point

When firms must make capital investments before meeting workers and wages are determined by ex post Nash bargaining, a holdup problem arises: firms underinvest because workers capture some of the return to capital in wage negotiations. Acemoglu and Shimer (1999) show this holdup is unavoidable under any “regular” bargaining game, but that wage posting with *directed search* resolves it: workers can observe capital levels and direct their search, creating Bertrand competition among vacancies that internalises the ex ante investment externality and achieves constrained efficiency.

Model structure

Continuous time. A continuum of risk-neutral workers and a larger continuum of risk-neutral firms. Firms invest in capital $k > 0$ at marginal cost p before posting a vacancy. Output of a filled job: $f(k)$, strictly increasing and concave with Inada conditions. Capital depreciates at rate s (matches also break at rate s).

- **Matching technology:** CRS matching function. Queue length Q (unemployed workers per vacancy) endogenous. Worker match rate $\mu(Q)$, vacancy fill rate $\eta(Q) = Q\mu(Q)$.
- **Ex post Nash bargaining (baseline):** Worker power β . Equilibrium investment k^B satisfies:

$$\frac{(1 - \beta) \eta(Q)}{r + s + (1 - \beta) \eta(Q)} \cdot \frac{f'(k^B)}{r + s} = p.$$

Holdup: Under any regular bargaining game, $k^B < k^S$ (social optimum) unless $\beta = 0$; if $\beta = 0$ firms earn excess returns, causing excessive vacancy entry. No single β achieves both conditions simultaneously.

- **Wage posting with directed search:** Firms post wages w in advance; workers observe posted wages and direct their search, endogenising $Q(w)$. Firms choose w to maximise expected profit, choosing capital k jointly. The optimal wage posting commitment exactly solves both efficiency conditions simultaneously.
- **Limited information robustness:** Efficiency holds even when each worker observes only *two* wage offers: Bertrand competition between two vacancies is sufficient.

Key findings

1. With ex post Nash bargaining, the equilibrium is always inefficient regardless of β .
2. Wage posting with directed search achieves constrained efficiency.
3. Full information is not required; observing two wage offers suffices for efficiency.

4. The key channel is workers' ability to *direct* their search toward capital-intensive jobs — not wage posting per se.

Takeaways for your project

1. **Training as a pre-match investment.** In your model, the decision to train and enter market S is a pre-match investment analogous to capital investment k here. Acemoglu and Shimer (1999) show that ex post Nash bargaining causes underinvestment in such pre-match capital, providing the theoretical foundation for why the decentralised training threshold $\bar{x}(z)$ is too high relative to the social optimum.
2. **Why CPR over pure Nash bargaining.** Your skilled market uses CPR Nash bargaining with Bertrand competition from outside offers — a partial internalisation of investment returns. Acemoglu and Shimer (1999) show pure Nash bargaining (without outside-offer competition) is always inefficient. The CPR mechanism moves your model toward efficiency for *within-market* investments; the training margin (between-market) remains distorted.
3. **Policy justification for training subsidies.** Acemoglu–Shimer's holdup directly justifies your Policy Exercise (2): if workers underinvest in training because trained wages are insufficient relative to costs (due to Nash bargaining extracting share β), then flow training subsidies or upfront cost reductions that lower $\bar{x}(z)$ are efficiency-improving.
4. **Random vs. directed search.** Your model uses random search (standard DMP). Acemoglu and Shimer (1999) show directed search achieves the efficient allocation. This is an important alternative specification: under directed search, workers would direct themselves toward the skilled market when training subsidies are introduced, generating faster adjustment. The efficiency properties of your model depend on whether search is directed or random — an area for discussion in the welfare analysis.
5. **Upfront vs. flow training costs and the holdup.** If workers pay training costs before matching (upfront cost), the holdup is most severe — the full training investment is sunk before bargaining. If costs are ongoing while employed (flow subsidy scenario), the holdup is partially alleviated. This directly explains why your Policy Exercise (2) distinguishes between upfront cost reductions and flow subsidies: the two instruments correct different parts of the investment distortion.

22 Audoly et al. (2022) — “Job Ladder, Human Capital, and the Cost of Job Loss”

Main point

Audoly et al. (2022) build and estimate a rich structural search model combining a job ladder, firm-specific human capital, and general human capital accumulation to decompose

the large and persistent earnings losses suffered by high-tenure displaced workers. Job ladder effects (falling to a less productive employer) account for 48–56% of cumulative wage losses, while firm-specific human capital loss accounts for 28–37%, with general skill depreciation playing a smaller role.

Model structure

Continuous-time equilibrium search model:

- **Heterogeneous firms:** Firms draw permanent productivity p from a fixed distribution. Workers sort into more productive jobs over time (job ladder).
- **On-the-job search:** Unemployed workers receive offers at rate λ_0 ; employed at rate λ_1 . Workers accept outside offers that improve their position.
- **Wage setting:** Sequential auction protocol (Postel-Vinay–Turon 2010, extending CPR 2006). Wages are renegotiated only when a credible threat exists: an outside offer or a match-specific productivity shock. This generates wage stickiness within jobs.
- **General human capital:** h_g accumulates at rate ρ_g while employed, depreciates at rate δ_g while unemployed.
- **Firm-specific human capital:** $h_s(s)$ increases with tenure s at the current employer at rate ρ_s ; destroyed completely upon separation.
- **Endogenous job destruction:** Match-specific productivity hit by i.i.d. shocks at Poisson rate μ ; match destroyed if new productivity falls below the endogenous reservation.
- **Estimation:** Indirect inference on longitudinal German Social Security data (IAB), targeting tenure-wage profiles, job-switching rates, and post-displacement wage losses.

Key findings

1. Job ladder losses account for 48–56% of cumulative post-displacement wage losses.
2. Firm-specific human capital loss accounts for 28–37%.
3. General human capital depreciation during unemployment accounts for the remainder (~ 15 –20%).
4. Without controlling for specific capital, studies attribute too much of wage losses to the job ladder — the “pure job ladder” model overstates this channel by 10–20 pp.
5. Most persistence in earnings losses comes from wages, not employment.

Takeaways for your project

1. **CPR wage protocol in the skilled market.** Audoly et al. directly use the CPR (2006) sequential auction as their wage-setting protocol. This is exactly your skilled market S wage mechanism. Their decomposition of wage losses into job ladder, firm-specific, and general HC provides a benchmark for what your job ladder produces in terms of wage dynamics.
2. **Job ladder quantitatively dominant.** Audoly et al. show that job ladder losses dominate displacement costs. This gives an empirical target for the magnitude of job ladder rents in your skilled sector — a key quantity for calibrating the CPR wage contract parameters.
3. **General HC depreciation calibration.** Audoly et al. estimate a non-trivial general HC depreciation rate during unemployment. Combined with Cohen et al. (2025)’s finding of zero measurable depreciation in cognitive skills, this suggests the depreciation in Audoly et al. may reflect other channels (re-matching risk, stigma). In your baseline calibration, setting the general HC depreciation rate near zero is consistent with Cohen et al. (2025).
4. **Within-skilled-market HC dynamics.** Your training threshold $\bar{x}(z)$ determines which workers enter market S and thus begin to accumulate both forms of HC on the job ladder. Audoly et al. provide the within-market HC dynamics that your model should be consistent with.
5. **Estimation by indirect inference.** Audoly et al.’s moment selection (tenure profiles, job-switching rates, displacement losses) provides a template for what moments are informative about HC accumulation parameters — relevant for your SMM strategy.

23 Cohen et al. (2025) — “Skill Depreciation during Unemployment”

Main point

Using panel data on German workers surveyed at the onset of unemployment and at three points over the following three years — with measured cognitive and non-cognitive skills linked to administrative records — Cohen et al. (2025) find *no evidence* that general skills decline during unemployment spells, despite the fact that reemployment hazard rates and wages fall sharply with unemployment duration. The paper concludes that general human capital depreciation is unlikely to be the primary explanation for duration dependence.

Identification strategy and key magnitudes

- **Data:** Panel drawn from the SOEP (Germany), supplemented with German Social Security administrative data. Skills measured at onset of unemployment and at 6, 12, and 36 month follow-ups. Cognitive battery: math, verbal fluency, immediate and

recall memory. Non-cognitive battery: conscientiousness, locus-of-control, patience, reciprocity, risk tolerance, social trust. U.S. replication panel confirms results.

- **Identification:** Within-worker panel design (holding unobserved worker characteristics constant); difference-in-differences comparing workers still unemployed at follow-up to early exiters.
- **Key magnitudes:** (i) Reemployment hazard is **5 times** higher right after job loss than after 2 years; (ii) reemployment wages fall by **20%** for long-term vs. short-term unemployed; (iii) yet measured cognitive and non-cognitive skills show **zero trend** over up to 3 years of unemployment.
- Skills at onset predict prior wages with $R^2 = 5\text{--}8\%$, confirming economic relevance.

Key findings

1. No measurable depreciation in any of 10+ cognitive and non-cognitive skills over up to 3 years of unemployment.
2. Both job-finding rates and reemployment wages decline substantially with duration — duration dependence exists.
3. The two findings together imply general HC depreciation is *not* the primary driver of duration dependence.
4. Depression and loneliness indicators rise significantly, suggesting psychological deterioration without corresponding skill loss.
5. The null result holds for Germany and the United States.

Takeaways for your project

1. **HC depreciation calibration.** Your model should set the general HC depreciation rate during unemployment near zero in the baseline calibration, consistent with Cohen et al. (2025). If Audoly et al. (2022)’s estimated depreciation rate is non-trivial, the discrepancy must be reconciled (likely through non-HC channels such as stigma or discrimination).
2. **Duration dependence via non-HC mechanisms.** Your model generates duration dependence through declining match quality or aggregate state z shifting. Cohen et al. (2025)’s evidence validates not incorporating explicit HC depreciation during unemployment to explain duration patterns.
3. **Training subsidies and skill maintenance.** Cohen et al. (2025) implies flow training subsidies need not be justified by preventing skill deterioration during unemployment. The welfare justification in your model is instead the correction of the hold-up (per Acemoglu and Shimer (1999)): subsidies lower $\bar{x}(z)$ and improve market- S access, not skill maintenance per se.

4. **Higher b_S and duration.** Your Policy Exercise (1) raises b_S (skilled workers’ outside option). Cohen et al. (2025) implies that longer skilled unemployment spells (from higher b_S raising selectivity) do not impair workers’ human capital. The selectivity channel (better match quality) operates cleanly without a skill-depreciation cost.

24 Dellas and Sakellaris (2003) — “On the Cyclicity of Schooling”

Main point

Dellas and Sakellaris (2003) provide a theoretical model showing that opportunity cost effects dominate ability-to-pay effects in the education investment decision, making college enrolment countercyclical in the absence of binding borrowing constraints. Using the October CPS for U.S. 18–22-year-olds over 1968–1988, they confirm that a one percentage point increase in the unemployment rate is associated with approximately a 2% increase in college enrolment, with cyclical recessions generating swings of over 200,000 students.

Specification and key magnitudes

Discrete-time OLG utility maximisation. Individual j chooses to enrol in period t if the net gain $\Omega_t(j)$ is positive, where:

- The direct cost $Q_t(j) = Q_t + g(j)$ with $g'(j) < 0$ (higher ability j has lower cost);
- $\partial\Omega_t/\partial w_{Nt} < 0$ (higher unskilled wage reduces enrolment incentive);
- $\partial\Omega_t/\partial \mathbb{E}_t(w_{t+1} - w_{Nt+1}) > 0$ (higher expected skill premium raises enrolment).

Empirics: probit/logit of individual enrolment on state unemployment rate (with controls for family income, age, year fixed effects). **Key magnitude:** 1 pp increase in unemployment rate $\rightarrow \sim 2\%$ increase in college enrolment rate.

Key findings

1. College enrolment is strongly countercyclical in the United States.
2. The opportunity cost (forgone low-skill wages while in school) dominates the ability-to-pay effect (credit constraints).
3. For enrolment to be procyclical, credit markets would have to be strongly procyclical *and* credit-constrained students would have to be predominantly high-ability — both empirically implausible.

Takeaways for your project

1. **Countercyclical training as the baseline prediction.** In your model, the training threshold $\bar{x}(z)$ adjusts with aggregate state z . During recessions (bad z), low-skill wages and job-finding fall, reducing the opportunity cost of training, shifting $\bar{x}(z)$ downward (more workers train). Dellas and Sakellaris (2003) identify this mechanism.
2. **SMM moment.** The countercyclical training decision is an empirical moment you can target in SMM. The magnitude ($\sim 2\%$ enrolment increase per pp of unemployment) provides a calibration target for the responsiveness of $\bar{x}(z)$ to aggregate conditions.
3. **No borrowing constraints needed.** Dellas and Sakellaris (2003) show countercyclical enrolment is robust without explicit credit constraints, validating your baseline model that abstracts from credit markets.

25 Clark (2011) — “Do Recessions Keep Students in School?”

Main point

Clark (2011) exploit 30 years of regional panel data for England (1975–2005) to show that youth labour market conditions have large effects on post-compulsory enrolment. A 1 pp increase in youth unemployment raises enrolment by approximately 0.4–0.9 percentage points (twice as large as prior UK time-series estimates), which explains why enrolment stagnated during the long mid-1990s economic expansion.

Identification strategy

Panel regression with region and year fixed effects, first-differenced to address unit roots, with region-specific trends. Identification exploits differential changes in youth unemployment across 8 English regions over time (e.g., North vs. South East during manufacturing vs. service recessions). Youth unemployment measured among 18–19-year-olds to avoid reverse causation from 16-year-old enrolment decisions. Parental unemployment (income effect) controlled separately: the income effect is negative and smaller than the opportunity cost effect, so the net estimate is a lower bound on the opportunity cost.

Key findings

1. $\hat{\beta}_1 \approx 0.4\text{--}0.9$ (pp enrolment per pp youth unemployment).
2. Effects at least twice as large as prior UK time-series estimates.
3. Stagnation of enrolment after mid-1990s largely explained by the long UK expansion.

4. Income effects (parental unemployment) work in the opposite direction but are smaller in magnitude.
5. Opportunity cost channel confirmed in a well-identified regional panel setting.

Takeaways for your project

1. **Corroboration across countries.** Together with Dellas and Sakellaris (2003), Clark’s regional panel evidence confirms countercyclical training using a different identification strategy and different country. These two papers together give strong support for the cyclical mechanism in your model.
2. **SMM calibration target.** Clark’s $\hat{\beta}_1 \approx 0.4\text{--}0.9$ (UK) and Dellas–Sakellaris’s $\sim 2\%$ (U.S.) both bound the elasticity of training entry with respect to unemployment — moments you can use to discipline $\bar{x}(z)$ ’s aggregate-state responsiveness.

26 Barr and Turner (2015) — “Out of Work and Into School”

Main point

Barr and Turner (2015) provide the first causal evidence that UI benefit duration affects post-secondary enrolment. Exploiting the staggered rollout of UI extensions during the Great Recession, they find that an additional 10 weeks of UI benefits increases enrolment by approximately 1.8 percentage points ($\sim 20\%$), with effects concentrated in two-year institutions and larger in states with more permissive “approved training” rules.

Identification strategy

October CPS (2007–2012) merged with state-level UI parameters (benefit duration by state-year from Emergency Unemployment Compensation (EUC) and Extended Benefits (EB) triggers). Exploits quasi-experimental variation: states crossing pre-determined unemployment rate triggers received additional benefit tiers (up to 99 weeks total). Regression:

$$P(\text{Enrol}) = \alpha + \beta \cdot (\text{weeks of UI available}) + \text{controls}.$$

Key controls: state UR, demographics, year FE, state FE.

Key findings

1. 10 additional weeks of UI $\rightarrow$ +1.8 pp in enrolment probability ($\approx 20\%$ relative increase).
2. Effects primarily from two-year (community college) enrolment.
3. Effects larger in states where UI-eligible workers can pursue broad academic programmes.
4. Enrolment increase concentrated among workers aged 20–30.

Takeaways for your project

1. **Direct evidence for Policy Exercise (1).** Barr and Turner (2015) provide direct causal evidence that higher b_U (longer UI $\equiv$ higher outside option in the unskilled market) increases training enrolment. This supports the prediction of your model that raising b_U shifts $\bar{x}(z)$ downward (more workers train), at least for marginal workers.
2. **Enrolment elasticity as SMM target.** The 1.8 pp increase per 10 weeks gives an empirical elasticity of training entry with respect to b_U that you can use to validate your model’s training response.
3. **Composition of trainees.** Effects concentrated among 20–30-year-olds at two-year colleges suggest the marginal trainee is a low-ability displaced worker from the unskilled market — corresponding to workers with ability just above $\bar{x}(z)$ in your model.
4. **Approved training laws as a flow subsidy analog.** The moderation by approved training laws is analogous to your flow training subsidy scenario. Barr and Turner (2015) show these complementarities can be quantitatively large, motivating your Policy Exercise (2).

27 Bičáková et al. (2021) — “Caught in the Cycle”

Main point

Using 58 cohorts of UK college graduates (1957–2014 entry cohorts), Bičáková et al. (2021) find that male graduates who enrol during worse economic conditions earn systematically higher average wages later ($\sim 1.7\%$ per 3 pp rise in unemployment at enrolment). This is not explained by selection, conditions at graduation, field of study, or differential employment. The paper attributes the result to greater *effort* exerted during college by students who enrol in bad times.

Identification strategy and key magnitudes

Cohort-based wage regression:

$$w_{i,c,t} = \alpha_t + \gamma \cdot \text{age}_{i,t} + \delta \cdot U_c + \text{trend}(c) + \varepsilon_{i,c,t}$$

where U_c is the national unemployment rate in the year of enrolment and identification exploits deviations from a quadratic cohort trend. Within-cohort regional variation provides additional identification. **Key magnitude:** 3 pp rise in enrolment-year unemployment $\rightarrow$ +1.7% in average male cohort wages (effect persists into mid-career).

Mechanisms ruled out

Selection into employment (no differential full-time employment); conditions at graduation (enrolment and graduation U are positively correlated, working against the result); major shifts (no large reallocation to higher-earning fields); high-school grades are similar or slightly worse for bad-cohort enrollees.

Mechanism supported

University grades are higher for bad-enrolment cohorts despite similar high-school performance — consistent with increased effort during college.

Takeaways for your project

1. **Quality of recession-cohort trainees.** When $\bar{x}(z)$ falls during recessions (more workers train), the marginal trainee has lower ability — but Bičáková et al. (2021) show total enrolled cohorts exhibit higher effort/quality. This compositional effect complicates the prediction that recession cohorts are lower quality; it suggests an effort dimension your model abstracts from.
2. **Entry conditions affect long-run match quality.** The wage premium associated with recession-cohort training is an auxiliary moment to validate the model’s predictions about within-cohort wage dispersion in market S as a function of entry conditions z .
3. **Job ladder dynamics.** Workers trained in bad times who subsequently enter market S face a recovering skilled market and begin climbing the CPR job ladder. The combination of higher effort during training and recovery-phase job ladder dynamics may together explain the persistent wage premium.

28 Bičáková et al. (2023) — “Make Your Own Luck”

Main point

Replicating and extending the 2021 UK result to the United States using nearly 40 cohorts of American college graduates, Bičáková et al. (2023) find robust evidence that U.S. workers who enrol in college during high unemployment subsequently earn higher wages, particularly women. Up to one-third of the effect is explained by geographic sorting (migration to higher-paying states), but a substantial wage premium remains.

Identification strategy and key magnitudes

American Community Survey (ACS), 2009–2018, 40 enrolment cohorts. Exploits within-cohort variation in state-level unemployment rates:

$$w_{i,s,c} = \lambda_c + \mu_s + \beta \cdot U_{s,c} + X_i' \gamma + \varepsilon_{i,s,c}$$

where $U_{s,c}$ is the state unemployment rate at enrolment for cohort c born in state s . **Key magnitude:** 1 pp higher state UR at enrolment $\rightarrow$ +0.6–1% in annual earnings (women). Up to one-third of the effect explained by geographic sorting.

Key findings

1. U.S. college graduates who enrolled during high unemployment earn higher subsequent wages, particularly women.
2. One-third of the effect explained by geographic sorting; two-thirds unexplained by observable channels.
3. Consistent with behavioural changes (effort, ambition) induced by adverse economic conditions at college entry.

Takeaways for your project

1. **Cross-country consistency.** The 2023 paper confirms the 2021 UK finding in U.S. data, giving a cross-country consistent empirical pattern on training quality as a function of entry conditions z .
2. **Non-spatial residual.** Since your model has no spatial dimension, the relevant comparison is the two-thirds residual attributable to non-geographic channels. This non-spatial premium provides the more direct calibration target for your within-cohort skilled-market wage dynamics.

29 Blom et al. (2021) — “Investment over the Business Cycle”

Main point

Blom et al. (2021) show that cohorts exposed to higher unemployment during typical schooling years select more remunerative majors (engineering, accounting, business, nursing) and abandon low-earning fields (education, sociology). Using 50+ cohorts of U.S. college graduates and ACS data, a 1 pp increase in unemployment during schooling years induces a 2.9–4.2 pp reallocation of majors among completers, shifting expected graduate earnings upward.

Identification strategy and key magnitudes

De-trended multinomial logit (and linear approximation) on 4.8+ million observations from 10 ACS waves (2009–2018), covering 50+ birth cohorts:

$$s_{j,c} = \alpha_j + \beta_j \cdot U_c + f_j(c) + \varepsilon_{j,c}$$

where $s_{j,c}$ is the share of cohort c in major j , U_c is the national unemployment rate when the cohort turned 20, and $f_j(c)$ is a major-specific flexible trend. **Key magnitudes:** 1 pp increase in unemployment $\rightarrow$ 4.2 pp total major reallocation for women, 2.9 pp for men. Largest gainers: nursing, accounting, computer science (women); engineering, accounting, business (men).

Key findings

1. Recessions significantly alter the distribution of college majors toward higher-earning, better-employment-prospects fields.
2. Changes are economically consequential: graduates earn wages consistent with their new majors.
3. Both the extensive margin (enrolment) and intensive margin (major choice within college) respond to the business cycle.

Takeaways for your project

1. **Composition of skills within market S .** Blom et al. (2021) show that the *type* of skilled worker entering the labour market changes over the cycle. In your model, the skilled market receives a flow of workers who trained in aggregate state z . Blom et al. suggest these workers are not just quantitatively different ($\bar{x}(z)$ shifts) but qualitatively different (their skill composition shifts). Your model treats all trained workers as homogeneous above $\bar{x}$; this is a tractability restriction worth acknowledging.
2. **Intensive margin of investment.** Flow training subsidies (Policy Exercise 2) affect not just whether workers train but potentially what skills they acquire. Blom et al. (2021) show this intensive margin is quantitatively large (2.9–4.2 pp per pp unemployment), complementing the extensive-margin story in your model.
3. **Consistent with countercyclical quality improvement.** Combined with Bičáková et al. (2021, 2023), Blom et al. (2021) gives a consistent picture: during recessions, more workers train *and* they train in higher-quality skills, both entering market S with greater productivity. This amplifies the countercyclical quality-upgrading of the skilled labour pool.

30 Card et al. (2018) — “What Works? A Meta-Analysis of Recent Active Labor Market Program Evaluations”

Main point

Card et al. (2018) synthesise 207 active labour market programme (ALMP) evaluations with 857 programme estimates. Their main conclusions are: (i) ALMP impacts are near zero in the short run but grow to 5–12 percentage points in the longer run (2+ years); (ii) training programmes have larger delayed gains than job search assistance; (iii) impacts are larger for women and long-term unemployed; (iv) programmes work better in recessions. There is no significant publication bias.

Specification

Meta-regression on 207 studies published 1995–2014 from Europe, North America, and developing countries. Two estimation strategies: OLS of estimated programme effect on programme type, participant group, and time horizon; and ordered logit on sign/significance. Programme types: training, job search assistance (JSA), private sector employment subsidy, public sector employment subsidy, combined. Study design quality controlled for throughout.

Key findings

- Short-run employment effect: 1–3 pp on average, near zero.
- Medium-run (1–2 years): 3–5 pp.
- Long-run (2+ years): 5–12 pp.
- Training programmes: larger long-run gains than JSA.
- Recessionary timing amplifies programme effects.
- Female participants and long-term unemployed: systematically larger average effects.
- No evidence of publication bias.

Takeaways for your project

1. **Delayed gains from training.** Training programmes have delayed gains (medium to long run). This supports your interpretation of flow training subsidies: they improve the quality of workers entering market S , but benefits materialise through the job ladder over time. SMM moments should extend beyond the impact period.
2. **Cyclical complementarity.** Programmes are more effective in recessions — when aggregate state z is low, training interventions for unskilled workers are more valuable because job-finding in market U is harder (lower θ_U), raising the relative return to training. This validates your countercyclical $\bar{x}(z)$ and the cyclical value of training subsidies.
3. **Benchmark for policy magnitude.** A 5–12 pp long-run employment rate gain from training programmes gives a benchmark for validating the employment rate effects predicted by your training subsidy exercise.
4. **Training vs. JSA comparison.** Card et al. (2018)’s finding that training programmes outperform JSA over the long run suggests the content of the intervention (human capital formation vs. job matching) matters more than timing. This corroborates your distinction between upfront cost reductions (front-loading) and flow subsidies (sustained support): both outperform pure JSA-style interventions.

31 Lindner and Reizer (2020) — “Frontloading the Unemployment Benefit”

Main point

In November 2005, Hungary replaced a flat UI benefit path with a declining benefit profile while holding total benefit amount constant. Lindner and Reizer (2020) find that this “frontloading” reform reduced non-employment duration by 1.5 weeks, with no change in reemployment wages or job duration — a welfare improvement at zero fiscal cost. The result demonstrates that the benefit *path* (not just total generosity) affects job search behaviour.

Identification strategy

Interrupted time series design exploiting the November 2005 policy change. Workers entering unemployment just before receive the old flat benefit; those just after receive the new declining benefit. Total benefit entitlement is held constant: only the path changes, isolating the effect of benefit timing from the level effect. Hungarian administrative register data covering the full population of UI recipients.

Key findings

1. Non-employment duration fell by 1.5 weeks after the reform.
2. Reemployment wages: unchanged.
3. New job duration: unchanged.
4. The reform is revenue neutral (faster job finding offsets higher initial benefits).
5. Workers respond to the time *profile*, not just the level, of benefits.

Takeaways for your project

1. **Benefit path as a policy instrument.** Lindner and Reizer (2020) show that even holding total benefit generosity constant, the timing profile affects search behaviour. In your model, the policy exercise (1) raises b_U (the level). Lindner and Reizer (2020) suggest the path of b_U over the unemployment spell is an additional policy dimension — holding total benefit constant while raising early benefit to encourage training decisions before workers settle in market U .
2. **Match quality not affected by search duration.** Faster job finding from frontloading does not reduce reemployment wages. This is consistent with Nekoei and Weber (2017)’s finding that match quality is determined by the selectivity margin, not duration per se. Together, these papers support your modelling of match quality as determined by the training decision $\bar{x}(z)$, not unemployment duration.

3. **Revenue neutrality.** Your policy exercises should address fiscal neutrality. Lindner and Reizer (2020)’s revenue-neutral result suggests policies that accelerate transitions ($U \rightarrow S$ via training) may pay for themselves if enhanced productivity in market S generates higher tax revenue — a welfare calculation your framework can implement.
4. **Flow of b_U vs. upfront cost reduction.** The distinction between frontloaded benefits (higher early b_U) vs. constant benefits maps onto your Policy Exercise (2): flow training subsidies change marginal incentives at each period; upfront cost reductions change the initial investment calculus. Lindner and Reizer (2020) show frontloading is effective and welfare-positive.

32 Ulvestad and Skjelbred (2023) — “Escaping into a Master’s Degree in Times of Crisis?”

Main point

Using Norwegian register data on all bachelor’s graduates from 2007–2021, Ulvestad and Skjelbred (2023) show that master’s degree enrolment is strongly countercyclical: a one percentage-point increase in the field-specific unemployment rate raises applications by 6.5 pp and enrolment by 3.9 pp. A survey of Covid-19-era graduates documents that recession-driven enrollees differ systematically from those whose decision was not driven by labour market conditions.

Specification and key magnitudes

Panel regression exploiting cross-field, within-year variation in unemployment rates faced by recent graduates of each field of study (capturing the opportunity cost each type of worker faces). The survey component identifies behavioural heterogeneity between “crisis enrollees” (who would not have enrolled in a normal economy) and “regular enrollees.”

Key magnitudes:

- 1 pp rise in field-specific UR $\rightarrow$ +6.5 pp in master’s applications and +3.9 pp in master’s enrolment.
- Effects are concentrated among fields with high labour market exposure (business, social sciences).
- Crisis enrollees differ in motivation, effort, and post-graduation labour market expectations relative to regular enrollees.

Key findings

1. Master’s degree enrolment in Norway is strongly countercyclical, with large elasticities relative to U.S. and UK studies.

2. The opportunity cost mechanism dominates: when field-specific labour markets are weak, continued education becomes relatively more attractive.
3. Crisis enrollees are behaviourally distinct — a selection into training that is partly driven by pessimism rather than expected human capital returns.

Takeaways for your project

1. **European evidence for countercyclical training.** Together with Dellas and Sakellaris (2003); Clark (2011); Barr and Turner (2015), this paper gives a fourth country-setting confirming countercyclical training enrolment. The large Norwegian elasticities (3.9 pp per pp) are especially striking and support the quantitative significance of the opportunity cost channel in your model.
2. **Marginal entrant heterogeneity.** Crisis enrollees differ from regular enrollees. In your model, when $\bar{x}(z)$ falls during recessions, the marginal trainee has lower ability (just above the new lower $\bar{x}$) and may have different within-training effort profiles. Ulvestad–Skjelbred’s survey evidence motivates allowing for marginal-entrant heterogeneity in the quality of trained workers entering market S .
3. **Field-specific opportunity costs.** The paper uses field-specific unemployment rates, which corresponds to market-specific tightness ($\theta_U(z)$) in your model. This supports your modelling choice of separate market tightness for each skill group.

33 Autor, Dubé, and McGrew (2023) — “The Unexpected Compression”

Main point

Autor et al. (2023) document a striking post-pandemic compression of the U.S. wage distribution (2019–2023), in which wages at the 10th percentile rose sharply relative to the median and 90th percentile, partially reversing four decades of rising inequality. Using CPS data and cross-state variation in post-pandemic labour market tightness, they show that wage compression is concentrated among young, non-college workers and is driven by job-to-job reallocation from lower-paying to higher-paying firms along the job ladder. The theoretical lens is a canonical Burdett–Mortensen (1998) job ladder model: tighter markets reduce employer monopsony power (raise the wage–separation elasticity), spurring upward wage growth.

Specification and key magnitudes

Cross-state variation in pandemic labour market tightness as the key source of identification. Wages at the 10th percentile relative to median (and 90th) decomposed by education group and age cohort. The Burdett–Mortensen framework: when the market tightens, workers

are more likely to receive outside offers and move up the job ladder, compressing the cross-sectional wage distribution by “dragging up” workers at the bottom of the ladder.

Key magnitudes:

- Non-college workers aged under 35: largest relative wage gains post-pandemic.
- Wage compression concentrated in states with larger post-pandemic tightening.
- Up to half of the long-run inequality increase since 1980 reversed during 2019–2023 at the lower end of the distribution.

Takeaways for your project (peripheral Family 3 reference)

1. **Empirical validation of the job ladder in the unskilled market.** Your model’s unskilled market uses standard Nash bargaining (no CPR job ladder), but Autor et al. (2023) apply a Burdett–Mortensen job ladder explicitly to the *low-wage* segment. If you add a job ladder to market U in extensions, Autor et al. (2023) provides the empirical motivation and identification strategy.
2. **Aggregate-state effects on wage distribution.** Autor et al. (2023)’s result that aggregate tightening (z improving) compresses the lower-end wage distribution through job reallocation is consistent with your model’s prediction that $\theta_U(z)$ rises in booms, improving match quality in market U . The aggregate-state switching in your model generates analogous cyclical wage distribution dynamics.
3. **Outside options and wage bargaining.** The mechanism in Autor et al. (2023) — tightness reduces monopsony power, raises wages — is the same as your policy exercise (1): raising b_U improves unskilled workers’ outside option, compressing the match-quality distribution upward through the DMP reservation-quality channel p_U^* . Autor et al. (2023) provides reduced-form empirical grounding for this channel.

Note: Autor et al. (2023) is circulated as an NBER Working Paper (No. 31010, 2023; revised 2024) and is not yet published in a refereed journal. It can be cited as a working paper with appropriate caveat.

Part II

Literature Families

Below are the papers found, grouped by family, with publication status noted. Papers marked [WP] are working papers and are included for awareness only — per your instruction, they should not be cited.

34 Family 1: Education Choice in Search Models

These are your most direct predecessors. They introduce search frictions into the education decision but do not close both gaps (simultaneous frictional equilibrium in both markets + job ladder in the skilled market + aggregate state transitions).

Paper	Journal	Key idea	Gap your paper fills
Charlot and De-creuse (2005)	<i>Labour Economics</i>	Two markets in simultaneous frictional equilibrium, endogenous composition via ability threshold, Nash bargaining, binary education choice	No job ladder in skilled market, no aggregate states, ability-independent schooling cost
Charlot et al. (2005)	<i>European Economic Review</i>	Adaptability and productivity as two margins of education return in a single-technology-segmented market	Single market (no skilled/unskilled segmentation), no binary threshold, no job ladder, no aggregate states
Wasmer (2006)	<i>AER</i>	General vs. specific human capital in a single frictional market with firing costs; endogenous job destruction	Single market only (no two-market segmentation at all), no composition externality, no job ladder

Paper		Journal		Key idea	Gap your paper fills
Charlot and De-crease (2010)		<i>Labour nomics</i>	<i>Eco-</i>	Two-dimensional heterogeneity (ability + cost), simultaneous two-market equilibrium, over-education for rich / under-education for poor	No job ladder in skilled market, no aggregate states
Decreuse and Granier (2013)		<i>Labour nomics</i>	<i>Eco-</i>	Education and hours, joint determination	Same limitations
Charlot and Malherbet (2013)		<i>Labour nomics</i>	<i>Eco-</i>	Education + EPL + dual contracts (STJ/LTJ) + endogenous job destruction; hold-up under Nash bargaining causes underinvestment; direct education subsidy is first-best	Single market (no skilled/unskilled segmentation), no job ladder, no aggregate states
Agopsowicz (2016)		[WP]		Education choice + OJS	Working paper — do not cite

Your positioning. You are the first to combine: (1) binary education threshold splitting a heterogeneous population, (2) both markets in simultaneous frictional equilibrium, (3) a full job ladder (CPR 2006 style) in the skilled market, (4) endogenous job destruction in the unskilled market, and (5) aggregate state transitions. Charlot–Decreuse (2005) and (2010) have (1) and (2) but not (3)–(5). Wasmer (2006) has only a single market.

35 Family 2: Macro / GE Education Choice (Competitive)

These are the “big” education models in competitive frameworks. Your paper positions itself as complementary — bringing their logic into a frictional world.

Paper	Journal	Key idea	Gap your paper fills
Heckman et al. (1998)	<i>Review of Econ. Dynamics</i>	Skill formation, education choice, endogenous skill prices	Competitive markets, no search frictions or unemployment risk
Abbott et al. (2019)	<i>JPE</i>	OLG education choice, intergenerational transfers, GE	Same — competitive factor markets
Lee and Seshadri (2019)	<i>JPE</i>	Intergenerational transmission, education + human capital	Same
Bárány (2016)	<i>JoLE</i>	Education choice + directed technical change + minimum wage	Competitive markets, no search frictions

36 Family 3: Job-Ladder / Search-and-Matching Foundations

These are the models you build on technically.

Paper	Journal	Key idea	Your extension
Diamond (1982)	<i>RES</i>	Search equilibrium, Nash bargaining, bilateral surplus	Foundational building block
Pissarides (2000)	MIT Press	DMP: matching function, free entry, Beveridge curve, endogenous job destruction	Foundational; you add two-market structure and education margin
Postel-Vinay and Robin (2002)	<i>Econometrica</i>	Sequential wage competition, job-ladder model, two-sided heterogeneity	Extended to 2 markets with endogenous composition; Nash bargaining retained (CPR style)

Paper	Journal	Key idea	Your extension
Shimer (2005)	<i>AER</i>	Cyclical unemployment dynamics, Shimer puzzle	Motivates aggregate-shock analysis; your model resolves via heterogeneity amplification
Cahuc et al. (2006)	<i>Econometrica</i>	Game-theoretic foundation of sequential auction with Nash bargaining power	Direct foundation for your skilled-market wage equation
Robin (2011)	<i>Econometrica</i>	PVR with aggregate shocks, endogenous job destruction, amplification via worker heterogeneity	You add second market, education margin, and Nash bargaining throughout
Bagger et al. (2014)	<i>AER</i>	Tenure, experience, HC accumulation on the job ladder, piece-rate tractability	HC accumulation within market, not market-selection; methodological benchmark
Lise et al. (2016)	<i>REDyn</i>	Two-sided heterogeneity + complementarities + endogenous job destruction + OJS	Adds education margin; your model omits (x, p) complementarity
Lise and Robin (2017)	<i>AER</i>	Block-recursivity with aggregate shocks and two-sided heterogeneity; cone-shaped matching set	Take type distribution as exogenous within each market; you endogenise it via training threshold
Acemoglu (2001)	<i>JLE</i>	Good vs. bad jobs, hold-up distortion, policy effects on job composition	You add frictional two-market structure with endogenous education choice
Moscarini and Postel-Vinay (2024)	<i>Revue écon.</i>	OJS + business cycles, three amplification channels, employment distribution as state variable	Peripheral; relevant for cyclical dynamics in your skilled market

Paper	Journal	Key idea	Your extension
Autor, Dubé, and McGrew (2023)	NBER WP 31010 [WP]	Post-pandemic wage compression via job ladder in the low-wage segment; aggregate tightening reduces monopsony power and compresses lower-end wages	Peripheral; no education margin; unskilled market; do not cite as published

37 Family 4: Human Capital in Search (Within-Market)

These study training/HC *within* a given market, not market selection.

Paper	Journal	Key idea	Distinction from yours
Acemoglu and Shimer (1999)	<i>IER</i>	Hold-up with search frictions; directed search achieves efficiency; Nash bargaining always inefficient for pre-match investment	Training is pre-match investment; directed search vs. random search is a key extension
Bagger et al. (2014)	<i>AER</i>	HC accumulation on job ladder	Within-market, no education margin
Audoly et al. (2022)	<i>FRTBNY Staff Reports</i>	Job-ladder losses, firm-specific and general HC, decomposition of displacement costs via CPR protocol	Within-market; no education margin; calibration benchmark for job ladder rents in market S
Cohen et al. (2025)	<i>AEJ: Applied</i>	No measurable general skill depreciation during unemployment; duration dependence not driven by HC loss	Null HC depreciation calibrates skill-depreciation rate near zero; duration dependence via other channels
Flinn et al. (2016)	[WP]	Worker-firm joint training decisions	Working paper

Paper	Journal	Key idea	Distinction from yours
Acabbi et al. (2024)	[WP]	Firm-dependent HC, cyclical sorting, hysteresis	Working paper — interesting but not citable

38 Family 5: Cyclical Enrollment

Reduced-form evidence that enrollment is countercyclical — your model provides the structural mechanism.

Paper	Journal	Key idea
Dellas and Sakellaris (2003)	<i>Oxford Economic Papers</i>	Countercyclical enrollment, four decades of US data
Clark (2011)	<i>Economica</i>	Youth unemployment → enrollment in England
Barr and Turner (2015)	<i>J. Public Economics</i>	Great Recession → enrollment at non-selective institutions; UI role
Bičáková et al. (2021)	<i>Economic Journal</i>	Graduates who enrol in recessions earn higher wages
Bičáková et al. (2023)	<i>Labour Economics</i>	Same authors, extended analysis
Blom et al. (2021)	<i>JPE</i>	Recession cohorts shift to higher-earning majors; both extensive and intensive margin respond to the cycle
Ulvestad and Skjelbred (2023)	<i>Economics of Education Review</i>	Countercyclical master's enrolment in Norway; 3.9 pp enrolment increase per pp field-specific UR
Leibing (2026)	[WP]	Recent evidence on enrollment cycles + outcomes

39 Family 6: Outside Options and Match Quality

Empirical and structural work on how outside options shape search outcomes.

Paper	Journal	Key idea
Nekoei and Weber (2017)	<i>AER</i>	UI extension improves match quality via selectivity
Card et al. (2018)	<i>JEEA</i>	Meta-analysis: training programmes work better medium-run
Lindner and Reizer (2020)	<i>AEJ: Applied</i>	Frontloading vs. flow UI — different search responses
Cohen et al. (2025)	<i>AEJ: Applied</i>	Skill depreciation during unemployment

Part III

Papers to Read and Search Strategies

40 Papers to Look At More Closely

Beyond what is already in the literature review, the following papers are worth reading or skimming:

1. **Barr and Turner (2015)** — They directly connect UI benefit design to enrollment timing during the Great Recession. Very relevant to your outside-option → training-threshold story.
2. **Bičáková et al. (2021); Bičáková et al. (2023)** — They find that students who enrol during recessions earn more later. Your model could rationalise this: workers who train during recessions are adversely selected on ability (lower $\bar{x}$ → more marginal workers train) but enter a recovering skilled market.
3. **Blom et al. (2021)** — Community college enrollment and local labour markets. Good micro evidence for your countercyclical-enrollment prediction.
4. **Cohen et al. (2025)** — Skill depreciation during unemployment. Relevant to why the duration cost of generous outside options matters.
5. **Lindner and Reizer (2020)** — Frontloading vs. flow UI. Directly relevant to your Policy Exercise (2) (flow training subsidies vs. upfront cost reductions).
6. **Card et al. (2018)** — Meta-analysis finding that training programmes work better in the medium run. Your model’s job-ladder dynamics in the skilled sector provide a structural explanation.

41 Manual Search Strategies

41.1 Strategy 1: Forward Citations from Charlot and Decreuse (2005)

Go to the Google Scholar page for Charlot and Decreuse (2005) and click “Cited by.” This will surface every paper that builds on the education-in-search tradition. Filter by “Since 2015” to find recent work. This is the most targeted way to find papers in your direct predecessor family.

41.2 Strategy 2: Forward Citations from Postel-Vinay and Robin (2002)

The PVR paper has thousands of citations. Search within them using Google Scholar’s search-within-citations feature:

- “Postel-Vinay Robin” + “education” or “training” or “human capital” or “skill choice”
- This surfaces anyone who extended PVR with an education/training dimension.

41.3 Strategy 3: NBER Working Paper Search

Go to <https://www.nber.org/papers> and search:

- “education choice” + “search frictions”
- “job ladder” + “education”
- “training” + “search model” + “equilibrium”
- “outside option” + “match quality”

NBER working papers that are good enough often end up in top journals. Even if you will not cite the WP itself, it tells you what is in the pipeline and who is working on related questions.

41.4 Strategy 4: Journal-Specific Searches

Search within specific top journals for recent papers you might have missed:

- ***Econometrica***: “education” + “search” (last 10 years)
- ***AER***: “training” + “frictional” or “search” (last 10 years)
- ***JPE***: “human capital” + “job ladder” (last 10 years)
- ***REStud***: “education” + “matching” or “search frictions” (last 10 years)
- ***Journal of Labor Economics***: “education choice” + “unemployment” (last 10 years)
- ***Labour Economics***: “education” + “search” + “equilibrium” (last 10 years)

41.5 Strategy 5: Search for “Education” in Search-Model Surveys

Look for recent survey articles on search-and-matching models:

- Rogerson, Shimer & Wright (2005, JEL) — the classic survey. Check what they say about education/human capital and follow references.
- Any recent *Handbook of Labor Economics* chapter on search models.
- Wright, Kircher, Julien, Guerrieri (2021, JEL) survey on directed search — check if education choice appears.

41.6 Strategy 6: Conference Programmes

Check recent programmes for:

- **SaM (Search and Matching) conference** — annual conference specifically on search models. Papers presented in 2023–2026 will include unpublished work on the frontier.
- **NBER Summer Institute (Labor Studies)** — programme available online.
- **CEPR/IZA workshops on Labour Economics** — programme available online.
- **EEA/ESEM annual meetings** — search the programme for “education” + “search” or “training” + “frictional.”

Even if the papers are working papers, knowing who is working on what helps you position.

41.7 Strategy 7: Keyword Combinations on Google Scholar

Try these specific queries:

- "endogenous education" "search frictions" "general equilibrium"
- "training decision" "job ladder"
- "skill choice" "frictional labor market"
- "education margin" "unemployment"
- "outside option" "reservation wage" "match quality"
- "countercyclical enrollment" structural model
- "education policy" "search and matching"

41.8 Strategy 8: Author-Based Search

These authors work at the intersection of education and search. Check their recent publications and working papers:

- **Olivier Charlot** and **Bruno Decreuse** — the education-in-search tradition
- **Fabien Postel-Vinay** — PVR extensions
- **Jeremy Lise** — sorting and search
- **Costas Meghir** — structural labour, education
- **Lance Lochner** — human capital, education
- **Attila Lindner** — UI design, outside options (also your advisor’s co-author on Lindner and Reizer (2020))

42 Assessment: Where You Stand in the Literature

Unique contribution. No published paper in a top journal simultaneously:

1. Has endogenous education/training choice in a search model, with a binary threshold that splits a heterogeneous population into two distinct labour pools.
2. Places both markets in frictional equilibrium simultaneously, with separate free-entry conditions, separate tightness levels, and separate composition that feeds back into job creation in each market.
3. Endogenises the composition of both pools through the training threshold $\bar{x}(z)$.
4. Builds the skilled market on the CPR (2006) job-ladder framework (Nash bargaining with on-the-job Bertrand competition and on-the-job search cutoff p_S^{oj}).
5. Endogenises job destruction in the unskilled market via a DMP-style reservation-quality rule p_U^* .
6. Introduces aggregate state transitions $z \in \mathcal{Z}$ that shift the full parameter constellation of both markets jointly, making the training threshold $\bar{x}(z)$ cyclically sensitive.
7. Estimates the model by Simulated Method of Moments on U.S. data.

Positioning relative to direct predecessors.

- **Charlot–Decreuse (2005)** has (1) and (2) but not (3)–(7). It has endogenous composition via a threshold σ^* and two markets in simultaneous equilibrium — but both markets use static Nash bargaining with no job ladder, no endogenous job destruction (only exogenous death), an ability-independent schooling cost, and no aggregate states. The key distinction is the absence of the job ladder and aggregate dynamics, *not* the absence of simultaneous equilibrium.
- **Charlot–Decreuse–Granier (2005)** has neither (2) nor the binary threshold: it is a representative-agent model in a single (technology-segmented) market. There is no composition externality across two distinct pools.
- **Wasmer (2006)** has a single labour market with no segmentation by skill level. The correct distinction from your paper is *not* “two markets not simultaneously in equilibrium” but rather “single market, no market segmentation at all.” Wasmer’s endogenous job destruction mechanism is useful background for your unskilled market, but the overall architecture is orthogonal.
- **Charlot–Decreuse (2010)** has (1) and (2) via two-dimensional heterogeneity, but again no job ladder, no aggregate states, and symmetric Nash bargaining in both markets. Your paper breaks this symmetry: Nash bargaining in the unskilled market, CPR sequential auction in the skilled market.

- **Charlot–Malherbet (2013):** the file in the workspace is “Education and Employment Protection,” *Labour Economics* 20 (2013), 3–23 — a different paper from what was previously expected, but directly relevant. It combines endogenous education choice with EPL in a DMP matching model, derives a hold-up problem under Nash bargaining (workers underinvest in education because they receive only fraction β of the return), and shows that direct education subsidies are the first-best policy. The key gap from your paper is that Charlot–Malherbet (2013) has a *single* undifferentiated labour market (no market segmentation by skill level), no job ladder, and no aggregate state transitions. The paper belongs to Family 1 and is a useful Family 1 predecessor for motivating the hold-up distortion that your training subsidy corrects.

Positioning relative to job-ladder foundations.

- The PVR extensions (Lise and Robin (2017); Bagger et al. (2014); Robin (2011); Moscarini and Postel-Vinay (2024)) have the job-ladder structure but no education margin. Your paper adds the education/training threshold to the job-ladder world — a dimension none of these papers consider.
- The macro education models (Bárány (2016); Abbott et al. (2019); Heckman et al. (1998)) have endogenous education choice and GE feedbacks but in competitive settings. Your paper brings their logic into a frictional world with an explicit job ladder.

Theoretical foundations of training underinvestment.

- Acemoglu and Shimer (1999) provides the theoretical backbone for why search frictions cause underinvestment in pre-match education: ex post Nash bargaining causes a hold-up, so workers invest less in training than the social optimum requires. This paper directly motivates your training subsidy policy exercises — both flow subsidies and upfront cost reductions are welfare-improving precisely because the decentralised equilibrium features $\bar{x}(z)$ *too high* relative to the planner’s optimum. Charlot and Malherbet (2013) independently confirm the same hold-up mechanism in a DMP model with dual EPL.
- The hold-up is *partially* mitigated in your skilled market by the CPR sequential auction: Bertrand competition from outside offers moves workers toward their outside option value and reduces the rent extracted by Nash bargaining at hire. The remaining distortion operates at the *between-market* training margin, where workers bear the full upfront cost before any wage negotiation occurs.

Risk areas.

- The Charlot and Decreuse (2005) papers are in *Labour Economics* and *European Economic Review*, not in the very top 5. This is actually good for you: it means the question has not been “solved” at the top level. But you need to engage carefully with their work and show what is genuinely new — in particular, that you are not simply adding aggregate shocks to a model that is otherwise isomorphic to Charlot–Decreuse (2005).

- The Acabbi et al. (2024) WP on cyclical sorting and hysteresis is conceptually close in spirit (aggregate shocks $\rightarrow$ composition effects $\rightarrow$ hysteresis), but they do not have an education margin. Worth tracking but not currently citable.
- Make sure to clearly distinguish from Wasmer (2006): that paper is in a top-5 journal, and referees will ask about it. The key distinction is that Wasmer has a single labour market with no skill-market segmentation — not “two markets not simultaneously in equilibrium.”
- **Charlot–Malherbet (2013)** is now confirmed as “Education and Employment Protection,” *Labour Economics* 20 (2013). It is a legitimate Family 1 predecessor (single-market DMP + endogenous education + hold-up). Engage with it in the literature review as a model that identifies the same hold-up problem but lacks market segmentation, a job ladder, and aggregate states.

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